RF COMMUNICATION RANGE CALCULATOR

Advanced Link Budget Analysis & Path Loss Propagation Modeling

Department of Electronics And Communication (Advanced Communication Technology)
CSMSS Chh. Shahu College of Engineering, Chh. Sambhajinagar

TEAM MEMBERS & ROLL NUMBERS:

1. Chaitnaya Shinde - AC2152 (Team Leader)
2. Dipak Belkar - AC2126
3. Lokesh Norlawar - AC2146
4. Kiran Sasane - AC2118
5. Shruti Shinde - AC2145

PROJECT GUIDE & STATUS:

Guide: Prof. Amruta Kadli Mam
Department: Electronics & Communication (ACT)

Current Progress: 85% COMPLETED

Presentation Deadline: 31st October 2026

Project Abstract & Core Objectives

RF Communication Range Calculator & Free-Space Attenuation Analysis

1. Executive Summary

Reliable wireless communication demands accurate estimation of maximum effective operating distances before hardware deployment. The RF Communication Range Calculator provides an interactive computational link-budget tool for microwave & sub-GHz links.

2. Primary Objectives

- Automate Friis Transmission equation and Free-Space Path Loss (FSPL) formulations.
- Calculate 1st Fresnel Zone obstacle clearances for line-of-sight assurance.
- Compare 433 MHz, 868 MHz, 915 MHz, and 2.4 GHz LoRa/RF transceivers.
- Build an intuitive web dashboard allowing faculty and students to simulate radio links.
- Provide predictive link margin indicators with rain, vegetation, and urban fades.

Key Innovation & Community Impact:

Prevents radio hardware failure in rural disaster telemetry, smart agriculture telemetry, and college campus IoT networks by calculating exact transmitter power and antenna gains.

Under the Supervision of: Prof. Amruta Kadli Mam | Department of E&ACT

Mathematical Modeling & RF Link Budget

Formulas implemented in the RF Range Calculator Engine

Free Space Path Loss (FSPL)

FSPL (dB) = 20 log10(d) + 20 log10(f) + 32.44

Where:

d = distance in kilometers (km)

f = frequency in megahertz (MHz)

Received Power (Pr) & Link Margin

Pr (dBm) = Pt + Gt + Gr - FSPL - Lmisc

Fade Margin (FM) = Pr - Receiver Sensitivity

Pt = Transmitter Power (dBm)

Gt, Gr = Tx and Rx Antenna Gains (dBi)

1st Fresnel Zone Radius

R1 (m) = 8.656 * sqrt(d / f_GHz)

Maintains minimum 60% clearance

to avoid multi-path diffraction and signal fade.

Essential for line-of-sight validation.

Maximum Range Inversion

d_max = 10 ^ (([Pt+Gt+Gr-S_rx-FM-32.44) - 20log10(f)] / 20)

Computes theoretical maximum range

instantly based on hardware transceiver specs.

Supports LoRa, ZigBee, Wi-Fi, and NRF24.

Student Researchers: Chaitnaya Shinde, Dipak Belkar, Lokesh Norlawar, Kiran Sasane, Shruti Shinde

Project Milestones & Development Progress

Current Progress: 85% Completed | Target Defense: 31st October 2026

X **Milestone 1: RF Link Budget & Path Loss Formulation**

Status: 100% Completed & Verified with standard ITU-R propagation tables.

X **Milestone 2: Antenna Gain & Sensitivity Modeling**

Status: 100% Completed with Dipole, Yagi, and Patch antenna options.

X **Milestone 3: Interactive Web-Based Calculator UI**

Status: 100% Completed - Real-time slider calculation and responsive UI.

X **Milestone 4: Multi-Frequency Terrain & Clutter Calibration**

Status: 100% Completed - Includes urban, suburban, and rural loss factors.

-> **Milestone 5: Final Defense, Documentation & PPT Presentation**

Status: Underway (Due 31st October 2026) before Prof. Amruta Kadli Mam.

OVERALL PROGRESS: 85% COMPLETED

Verified by Prof. Amruta Kadli Mam